1. Identity

Document	STAR-PKG-STAR-SPHERE-EN · v2.2 · 2026-11-02
Package	star-sphere
Version	0.1.1
Capability	star.geometry.sphere
Maturity	Released & Verified (RELEASED)
Python	⩾3.9
Source commit	dc09f61
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_sphere

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_sphere-0.1.1-py3-none-any.whl
```

FILE	SHA-256
star_sphere-0.1.1-py3-none-any.whl	ba9e082c040ee312f5ac0c041a1fe09ff1ed14bf34de074a859dbf3af57640c6
star_sphere.py	109d55b2ce7627f10b231c80276369061a59e12450ebbbef7507183668b9beb1
star_sphere-0.1.1-source.zip	b134cb9c82b46baab58cd28aaea6d1ad11d2621dc85339bc325dca05a67519a4

5. What it does, and its limits

Angular separation, position angle and offset point for directions on the sphere, accurate from 1e-12 deg to the antipode; agrees with ERFA and astropy to 3e-14 deg on 600 pairs and refuses out-of-range angles.

A sphere: for distances on the Earth's ellipsoid use `star_geodesic` (the difference reaches 0.5 %). No proper motion, aberration or refraction: the directions are taken as given.

- `separation_deg` in [0, 180] with the arc-tangent formula: nearby directions (1e-12 deg) and nearly antipodal ones keep their digits. Reproduces Meeus Example 17.a (32.7930 deg). Since 0.1.1 the differences of longitude and latitude are taken in degrees before any conversion, so a separation of 1e-12 deg is right to the last digits (0.1.0 was wrong in the sixth digit at 1e-9 deg and in the third at 1e-12 deg: update if you measure tiny angles).
- `position_angle_deg` in [0, 360), from North through East; `offset(lon, lat, position_angle, distance)` returns (longitude in [0, 360), latitude) and is the inverse of the two.
- Measured on 600 pairs (300 random, 150 down to 1e-6 deg apart, 150 down to 1e-6 deg from the antipode): separation within 6e-14 deg of ERFA and astropy, position angle within 1e-15 rad of displacement, offset points within 4e-13 deg of astropy.
- Every function returns finite floats in those ranges or raises `ValueError`: non-numeric, boolean, NaN or infinite input, latitude outside [-90, 90], longitude or position angle outside [-360, 360], distance outside [0, 180]. For coincident directions the position angle is undefined and returned as 0.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_sphere_0_1_1,
  author  = {Cavazzini, Andrea},
  title   = {{star-sphere}: Angles between directions on the sphere},
  version = {0.1.1},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_sphere/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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Related: space activities and export control · acceptable use · evidence and limits · licence texts